

MT PLEASANT, MI, USA

WMO: 725424

Lat: 43.622N

Lon: 84.737W

Elev: 755

StdP: 14.30

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 04885

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -0.4	(c) 3.5	(d) -8.0	(e) 3.7	(f) 1.6	(g) -3.5	(h) 4.7	(i) 6.3	(j) 21.1	(k) 24.6	(l) 18.9	(m) 26.5	(n) 3.3	(o) 240	(p) 0.516

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.9	89.7	72.1	86.2	70.2	83.1	68.5	75.4	84.2	73.5	81.4	71.9	79.4	9.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.9	125.4	79.0	71.6	119.9	77.7	69.8	112.8	76.2	39.4	84.4	37.6	80.8	36.1	78.2	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	16.8	15.0	-7.6	93.0	6.6	2.8	-12.3	94.9	-16.1	96.6	-19.8	98.1	-24.6	100.1
			-8.1	78.3	6.5	2.4	-12.8	80.1	-16.6	81.5	-20.2	82.8	-24.9	84.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.3	22.8	23.6	33.3	45.5	56.9	66.1	70.5	68.8	61.8	49.8	38.6	28.9
	DBStd	18.92	10.51	10.00	10.98	9.11	8.63	6.62	5.84	5.45	7.52	8.48	8.43	8.60
	HDD50	3458	842	740	532	191	28	0	0	0	4	112	353	656
	HDD65	6960	1307	1160	985	587	280	64	15	25	147	477	792	1121
	CDD50	2489	0	0	13	57	242	482	637	581	358	106	12	1
	CDD65	515	0	0	1	3	30	96	187	141	51	6	0	0
	CDH74	5181	0	0	13	54	359	1009	1892	1272	514	68	0	0
	CDH80	1460	0	0	2	7	88	286	612	333	123	9	0	0
(14) Wind	WSAvg	6.0	6.8	6.8	6.7	7.4	6.3	5.3	4.8	4.4	4.6	5.9	6.5	6.6
(15) Precipitation	PrecAvg	30.7	1.5	1.1	2.0	3.1	3.0	3.4	2.8	3.5	3.4	2.7	2.7	1.8
	PrecMax	43.9	3.7	2.6	5.3	7.1	7.1	7.2	6.1	10.2	15.4	7.1	5.3	4.8
	PrecMin	20.6	0.0	0.0	0.1	0.7	0.1	0.4	0.7	0.7	0.0	0.2	0.6	0.2
	PrecStd	4.7	0.9	0.7	1.1	1.5	1.7	1.7	1.5	2.0	2.5	1.5	1.2	1.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.7	56.4	73.1	80.8	87.2	90.9	93.1	90.9	90.0	81.1	68.0	56.8
		MCWB	48.6	50.3	60.0	61.2	68.8	72.6	73.7	72.8	70.8	68.2	57.6	52.8
	2%	DB	45.0	45.6	62.8	73.2	82.1	87.3	90.1	86.9	83.3	74.9	60.7	48.1
		MCWB	41.5	41.5	53.6	57.3	65.5	69.4	73.2	70.5	68.4	62.6	52.7	44.7
	5%	DB	40.3	40.7	55.0	69.1	78.3	82.8	86.3	83.7	80.3	69.7	55.4	43.5
		MCWB	37.5	36.8	48.3	54.6	63.2	68.2	71.1	69.0	66.8	60.5	49.3	40.6
	10%	DB	36.5	36.6	49.5	62.8	73.2	80.3	82.5	81.1	76.6	64.3	52.1	39.5
		MCWB	34.6	33.6	43.9	50.8	60.9	66.5	68.9	67.9	65.3	57.9	47.3	36.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.8	50.0	62.1	64.6	71.7	76.1	78.7	77.1	74.5	70.4	58.7	54.1
		MCDB	50.4	56.4	73.5	75.5	81.8	85.9	86.9	84.5	82.8	77.3	63.6	56.8
	2%	WB	41.6	42.2	55.4	59.8	68.8	73.2	76.2	74.4	72.1	66.7	54.5	45.8
		MCDB	44.5	45.4	59.9	69.6	77.5	81.5	85.1	81.9	78.5	71.6	59.2	47.1
	5%	WB	37.5	36.9	49.6	56.4	66.3	71.2	74.0	72.4	70.0	62.6	50.8	41.2
		MCDB	39.5	40.0	54.1	65.8	74.2	79.6	82.6	79.6	75.6	67.6	54.5	44.0
	10%	WB	34.7	33.9	43.4	52.5	63.4	69.2	72.2	70.5	67.7	58.4	47.2	37.3
		MCDB	36.6	36.3	48.2	61.6	70.2	76.2	79.8	77.2	74.5	63.6	51.8	39.3
(35) Mean Daily Temperature Range	MDBR		12.9	14.7	17.6	20.6	21.9	21.5	21.9	21.6	22.0	18.5	14.5	10.8
	5% DB	MCDBR	15.5	17.6	26.5	29.7	27.2	24.9	24.8	24.3	25.9	26.3	21.1	15.4
		MCWBR	13.4	13.9	17.7	15.7	13.1	11.4	10.9	10.6	12.4	15.2	15.1	13.5
	5% WB	MCDBR	14.6	16.2	24.3	26.0	23.3	20.3	20.5	19.9	19.9	22.0	19.6	15.2
		MCWBR	13.1	13.7	18.0	16.8	13.3	11.2	10.5	10.8	12.1	15.8	15.8	14.0
(40) Clear-Sky Solar Irradiance	taub		0.287	0.321	0.329	0.364	0.391	0.416	0.407	0.399	0.376	0.336	0.305	0.282
	taud		2.356	2.251	2.380	2.318	2.294	2.257	2.286	2.306	2.368	2.478	2.506	2.429
	Ebn at Noon		267	274	289	285	279	272	273	271	269	268	261	260
	Edh at Noon		25	33	33	39	41	43	41	39	34	27	22	21
(44) All-Sky Solar Radiation	RadAvg		432	714	1141	1447	1706	1912	1935	1643	1315	817	489	344
	RadStd		38	82	125	128	161	114	101	83	125	89	64	34

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (56 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page